```
  }
}
```

❶ The top level section is named SmtpSettings.

❷ The SmtpSettings section has four properties.

❸ One of the properties, Credentials, also has properties.

Now that you have the configuration information stored in the appSettings.json file, you need to access specific config values. In the next couple of sections, you will look at some ways to achieve that. The first method shows how to reference values using string-based keys. Afterward, you will take a strongly typed approach instead, which reduces the potential for typing errors.

14.2.2 Accessing configuration settings programmatically by key

As mentioned previously, configuration data is accessible via the `IConfiguration` service, which is one of the default services registered with the DI system at application startup. You will make the `IConfiguration` service available to the `ProductionEmailService` class by injecting it into the constructor. Then you will use it to access individual settings using the section and property names as keys. The section is specified as the first part of the key, and subsequent properties are referenced by separating them with colons (:). For example, the `Host` property of the `SmtpSettings` section is referenced using the key `SmtpSettings:Host`. Listing 14.11 shows the fully modified `ProductionEmailService`.

NOTE Because the colon is essentially a reserved character, you cannot include it in configuration keys. This precludes the use of fully qualified URLs as configuration keys, for example.